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GitHub Link	**https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO3**

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1 Case Study

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0405**

SLOT : D

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

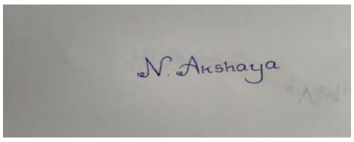
CO Weightage: 25% **Duration: 90 minutes**

Assessment Tool Weightage: 25% **Marks: 25**

Code of Conduct:

I **N.AKSHAYA(192425129)***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: **Name of the student: N.AKSHAYA**

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	*25 %*	*25%*	*20%*	*20 %*	*10 %*	**100%**
Q1						
Q2						
Total						

Signature of the Faculty: *Total Marks :*

QUESTIONS: 02 **Total Marks: 20**

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks) A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.

   Answer:

   1. Suitable CNN Architecture

   A suitable architecture for chest X-ray classification can be:

   Input Image (224 × 224 × 1)

   ↓

   Convolution Layer (32 filters, 3×3) + ReLU

   ↓

   Max Pooling (2×2)

   ↓

   Convolution Layer (64 filters, 3×3) + ReLU

   ↓

   Max Pooling (2×2)

   ↓

   Convolution Layer (128 filters, 3×3) + ReLU

   ↓

   Max Pooling (2×2)

   ↓

   Flatten

   ↓

   Fully Connected Layer (128 neurons) + ReLU

   ↓

   Dropout

   ↓

   Output Layer (2 neurons) + Softmax

   ↓

   Normal / Disease

   This architecture progressively learns simple features such as edges and textures and then more complex patterns representing abnormal regions.

2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.

   Answer:
   CNNs are suitable for medical image classification because they automatically learn spatial features directly from X-ray images, eliminating the need for manually designed features. The case specifically identifies **edges, textures, and abnormal regions** as important features learned by CNNs.

- **Input layer:** Receives the preprocessed chest X-ray image.
- **Convolutional layer:** Applies filters to detect visual patterns.
- **Filters:** Learn features such as edges, shapes, textures and disease-related patterns.
- **ReLU:** Introduces non-linearity and helps the network learn complex relationships.
- **Pooling layer:** Reduces feature-map dimensions and computational requirements while retaining important information.
- **Fully connected layer:** Combines extracted features for classification.
- **Softmax output:** Produces probabilities for the two classes: Normal and Disease.

3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.

   Answer:
   In a CNN, the same filter weights are applied across different locations of an image. Therefore, a filter does not need separate parameters for every pixel position.
   For example, instead of learning separate weights for every location, one **3 × 3 filter** can detect an edge wherever that edge appears.
   Therefore, parameter sharing:

- Reduces the number of trainable parameters.
- Reduces memory and computational requirements.
- Helps detect the same feature at different image locations.
- Makes CNNs more efficient than fully connected networks.

4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.

   Answer:
   Medical X-ray datasets can contain variations in **image size, brightness and image quality**, making overfitting an important concern.
   Data Augmentation:
   Images can be slightly rotated, shifted, scaled or brightness-adjusted to create additional training examples.
   Dropout:
   Randomly removes some neurons during training so that the network does not become overly dependent on particular neurons.
   Batch Normalization:
   Normalizes intermediate activations and can improve training stability and convergence.
   These techniques improve the model's ability to generalize to unseen X-ray images.

5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Answer:

Feature	**AlexNet**	**ResNet**
Architecture	Traditional deep CNN	CNN with residual/skip connections
Depth	Relatively shallow	Can be much deeper
Training	Easier for smaller networks	Designed to make very deep networks train effectively
Vanishing gradient	More susceptible in deeper versions	Skip connections help reduce the problem
Feature learning	Good	Stronger hierarchical feature learning
Computational cost	Generally lower	Generally higher depending on ResNet version
Medical image classification	Suitable baseline	Better for complex image features

Recommendation: ResNet
For medical X-ray classification, **ResNet is the more appropriate choice**, particularly when sufficient computational resources and training data are available. Its residual connections allow deeper networks to learn complex visual features while reducing the difficulty caused by vanishing gradients.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)
An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.

Answer:

1. CNN Architecture
For autonomous vehicle object recognition, the CNN can be designed as:
Camera Image
↓
Input Layer
↓
Convolution + ReLU
↓

Pooling
↓
Convolution + ReLU
↓
Pooling
↓
Deeper Convolution Layers
↓
Feature Representation
↓
Fully Connected / Detection Head
↓
Output
↓
Car | Pedestrian | Traffic Sign | Bicycle | Road Obstacle
The system must handle different **lighting conditions, viewing angles, distances and road environments**, so a deeper CNN with strong feature-learning capability is appropriate.

2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.

   Answer:
   Input Layer:
   Receives images captured by vehicle-mounted cameras.
   Convolution Layer:
   Extracts important spatial features from the image.
   Activation Layer:
   ReLU introduces non-linearity and allows the network to learn complex patterns.
   Pooling Layer:
   Reduces spatial dimensions while preserving important features.
   Fully Connected/Output Layer:
   Uses the learned representation to classify objects.
   CNN filters progressively learn:
   Low-level features → Mid-level features → High-level features
   For example:
   Edges → Shapes → Wheels/sign patterns/body shapes → Cars, pedestrians, bicycles, signs
   This progressive feature learning is one of the important concepts required in the case study.

3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.

   Answer:
   Parameter sharing means that the same convolutional filter is used at different locations in an image.

For example, a filter trained to identify a vertical edge can detect that edge whether it appears on the left, center or right side of the road image.
Therefore, parameter sharing:

1. Reduces the number of trainable parameters.
2. Reduces computational complexity.
3. Reduces memory requirements.
4. Enables detection of similar features at different locations.
5. Improves translation-related feature detection.

These benefits are particularly important for autonomous vehicles because objects can appear at different positions within camera frames.

4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.

Answer:
The model may overfit the training images and perform poorly on new road scenes.
Suitable techniques include:

- **Data augmentation:** Rotation, scaling, cropping and brightness changes.
- **Dropout:** Randomly disables neurons during training.
- **Batch normalization:** Stabilizes neural network training.
- **Early stopping:** Stops training when validation performance stops improving.
- **Weight regularization:** Penalizes excessively large weights.

These methods improve generalization to unseen driving environments. The assessment specifically expects discussion of overfitting and suitable regularization methods.

5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.

Answer:

Criteria	**AlexNet**	**ResNet**
Architecture	Conventional CNN	Residual CNN
Depth	Shallower	Much deeper possible
Parameter efficiency	Less efficient for deeper feature learning	Better feature reuse through residual connections
Training difficulty	Relatively straightforward	Easier to train very deep networks
Vanishing gradients	More vulnerable	Skip connections help overcome the problem
Feature learning	Good basic features	Strong hierarchical feature learning
Accuracy	Good baseline	Generally stronger for complex visual tasks
Real-time use	Lower computational requirement	Depends on ResNet version

6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

   Answer:
   ResNet is recommended for the autonomous vehicle application.
   The reasons are:

- It can learn complex hierarchical features.
- Residual connections help train deeper networks.
- It handles complex visual variations better.
- It can provide strong classification performance.
- It is more suitable for recognizing multiple object categories in challenging environments.

However, for a real-time autonomous vehicle system, the **specific ResNet variant should be selected according to available hardware and latency requirements**. A smaller ResNet can provide a better balance between accuracy and computational cost.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand